

SKILL 9:

```
This message is shown once a day. To disable it please create the
/home/nirvighna/.hushlogin file.
nirvighna@DESKTOP-VS57D2I:~$ cd ~/OS-Week1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l
total 96
-rw-r--r-- 1 nirvighna nirvighna 498 Sep 21 05:14 Makefile
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 27 16:59 bin
-rw-r--r-- 1 nirvighna nirvighna 54 Sep 25 15:03 input.txt
-rw-r--r-- 1 nirvighna nirvighna 1940 Sep 21 04:00 mini_shell.c
-rw----- 1 nirvighna nirvighna 1193 Sep 25 13:34 nano.2750.save
-rw-r--r-- 1 nirvighna nirvighna 54 Sep 25 17:34 output.txt
-rwxr-xr-x 1 nirvighna nirvighna 16440 Sep 25 18:02 producer_consumer
-rw-r--r-- 1 nirvighna nirvighna 1133 Sep 25 18:02 producer_consumer.c
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 27 16:58 src
-rw-r--r-- 1 nirvighna nirvighna 746 Sep 25 17:34 strace_output.txt
-rwxr-xr-x 1 nirvighna nirvighna 16352 Sep 25 17:48 wait_demo
-rw-r--r-- 1 nirvighna nirvighna 1382 Sep 25 17:47 wait_demo.c
-rwxr-xr-x 1 nirvighna nirvighna 16216 Sep 25 17:52 zombie
-rw-r--r-- 1 nirvighna nirvighna 625 Sep 25 17:51 zombie.c
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin
total 168
-rwxr-xr-x 1 nirvighna nirvighna 21208 Sep 27 16:59 builtin_shell
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 25 17:22 file_copy
-rwxr-xr-x 1 nirvighna nirvighna 19576 Sep 27 15:55 path_search
-rwxr-xr-x 1 nirvighna nirvighna 19784 Sep 21 05:34 practical1
-rwxr-xr-x 1 nirvighna nirvighna 18056 Sep 25 07:42 process_states
-rwxr-xr-x 1 nirvighna nirvighna 20560 Sep 27 12:34 session6_shell
-rwxr-xr-x 1 nirvighna nirvighna 20008 Sep 27 16:23 var_expand
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 27 15:48 waitpid_demo
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/builtin_shell
Built-in Dispatch Shell
Type 'help' for commands.
builtin-shell> |
```

```
builtin-shell> cd /abc/xyz
cd: No such file or directory
builtin-shell> cd /home/nirvighna/OS-Week1
builtin-shell> cd /temp
cd: No such file or directory
builtin-shell> cd -
cd: No such file or directory
builtin-shell> pwd
/home/nirvighna/OS-Week1
builtin-shell> |
```

```
builtin-shell> pwd
/tmp
builtin-shell> cd /home/nirvighna/OS-Week1
builtin-shell> pwd
/home/nirvighna/OS-Week1
builtin-shell> hello123
Invalid command: hello123
builtin-shell> |
```

```
builtin-shell> pwd
/home/nirvighna/OS-Week1
builtin-shell> cd /tmp
builtin-shell> pwd
/tmp
builtin-shell> cd /home/nirvighna/OS-Week1
builtin-shell> pwd
/home/nirvighna/OS-Week1
builtin-shell> exit
Shell exited.
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

SKILL 10:

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l src
total 52
-rw-r--r-- 1 nirvighna nirvighna 2780 Sep 27 16:58 builtin_shell.c
-rw-r--r-- 1 nirvighna nirvighna 1328 Sep 25 17:21 file_copy.c
-rw-r--r-- 1 nirvighna nirvighna 6065 Sep 21 04:44 history_shell.c
-rw-r--r-- 1 nirvighna nirvighna 1783 Sep 21 04:04 mini_shell.c
-rw-r--r-- 1 nirvighna nirvighna 3715 Sep 21 05:06 parser.c
-rw-r--r-- 1 nirvighna nirvighna 1054 Sep 27 15:54 path_search.c
-rw-r--r-- 1 nirvighna nirvighna 1702 Sep 21 05:32 practical1.c
-rw-r--r-- 1 nirvighna nirvighna 1702 Sep 21 05:29 practical.c
-rw-r--r-- 1 nirvighna nirvighna 1058 Sep 25 07:41 process_states.c
-rw-r--r-- 1 nirvighna nirvighna 2562 Sep 27 12:33 session6_shell.c
-rw-r--r-- 1 nirvighna nirvighna 2200 Sep 27 16:21 var_expand.c
-rw-r--r-- 1 nirvighna nirvighna 977 Sep 27 15:44 waitpid_demo.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin
total 168
-rwxr-xr-x 1 nirvighna nirvighna 21208 Sep 27 16:59 builtin_shell
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 25 17:22 file_copy
-rwxr-xr-x 1 nirvighna nirvighna 19576 Sep 27 15:55 path_search
-rwxr-xr-x 1 nirvighna nirvighna 19784 Sep 21 05:34 practical1
-rwxr-xr-x 1 nirvighna nirvighna 18056 Sep 25 07:42 process_states
-rwxr-xr-x 1 nirvighna nirvighna 20560 Sep 27 12:34 session6_shell
-rwxr-xr-x 1 nirvighna nirvighna 20008 Sep 27 16:23 var_expand
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 27 15:48 waitpid_demo
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cd ~/OS-Week1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ pwd
/home/nirvighna/OS-Week1
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin
total 168
-rwxr-xr-x 1 nirvighna nirvighna 21208 Sep 27 16:59 builtin_shell
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 25 17:22 file_copy
-rwxr-xr-x 1 nirvighna nirvighna 19576 Sep 27 15:55 path_search
-rwxr-xr-x 1 nirvighna nirvighna 19784 Sep 21 05:34 practical1
-rwxr-xr-x 1 nirvighna nirvighna 18056 Sep 25 07:42 process_states
-rwxr-xr-x 1 nirvighna nirvighna 20560 Sep 27 12:34 session6_shell
-rwxr-xr-x 1 nirvighna nirvighna 20008 Sep 27 16:23 var_expand
-rwxr-xr-x 1 nirvighna nirvighna 18080 Sep 27 15:48 waitpid_demo
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2  x  +  v  -  □  ×
GNU nano 7.2                  src/session6_shell.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

void parse_input(char *input, char *args[])
{
    int argc = 0;
    char *src = input;
    char *dst = input;

    while (*src != '\0' && argc < MAX_ARGS - 1)
    {
        while (*src == ' ' || *src == '\t' || *src == '\n')
            src++;

        if (*src == '\0')
            break;

        args[argc++] = dst;

        int escaped = 0;
        while (*src != '\0')
        {
            if (escaped)
            {
                *dst++ = *src++;
                escaped = 0;
            }
            else if (*src == '\\')
            {
                [ Read 133 lines ]
            }
        }
    }
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ nano src/session6_shell.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ cp src/session6_shell.c src/session6_s
hell_backup.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l src/session6_shell*
-rw-r--r-- 1 nirvighna nirvighna 2562 Sep 27 12:33 src/session6_shell.c
-rw-r--r-- 1 nirvighna nirvighna 2562 Sep 28 03:46 src/session6_shell_backup
.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l src/session6_shell*
-rw-r--r-- 1 nirvighna nirvighna 2562 Sep 27 12:33 src/session6_shell.c
-rw-r--r-- 1 nirvighna nirvighna 2562 Sep 28 03:46 src/session6_shell_backup
```